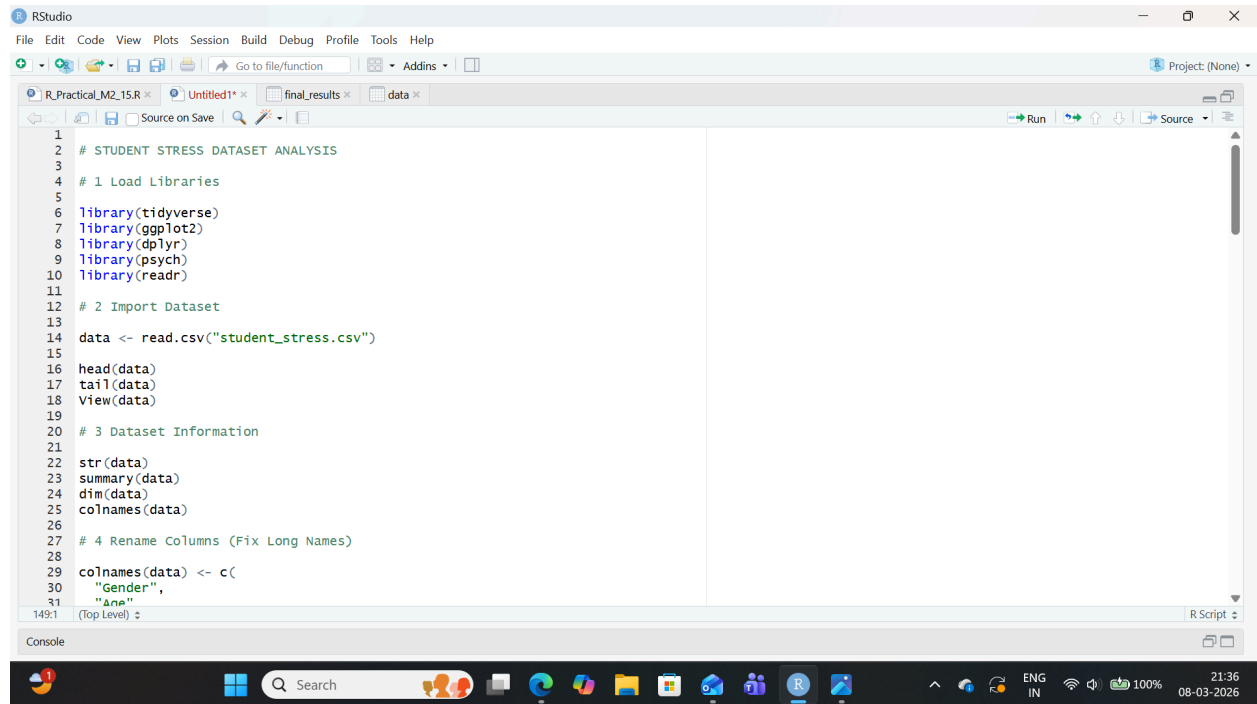
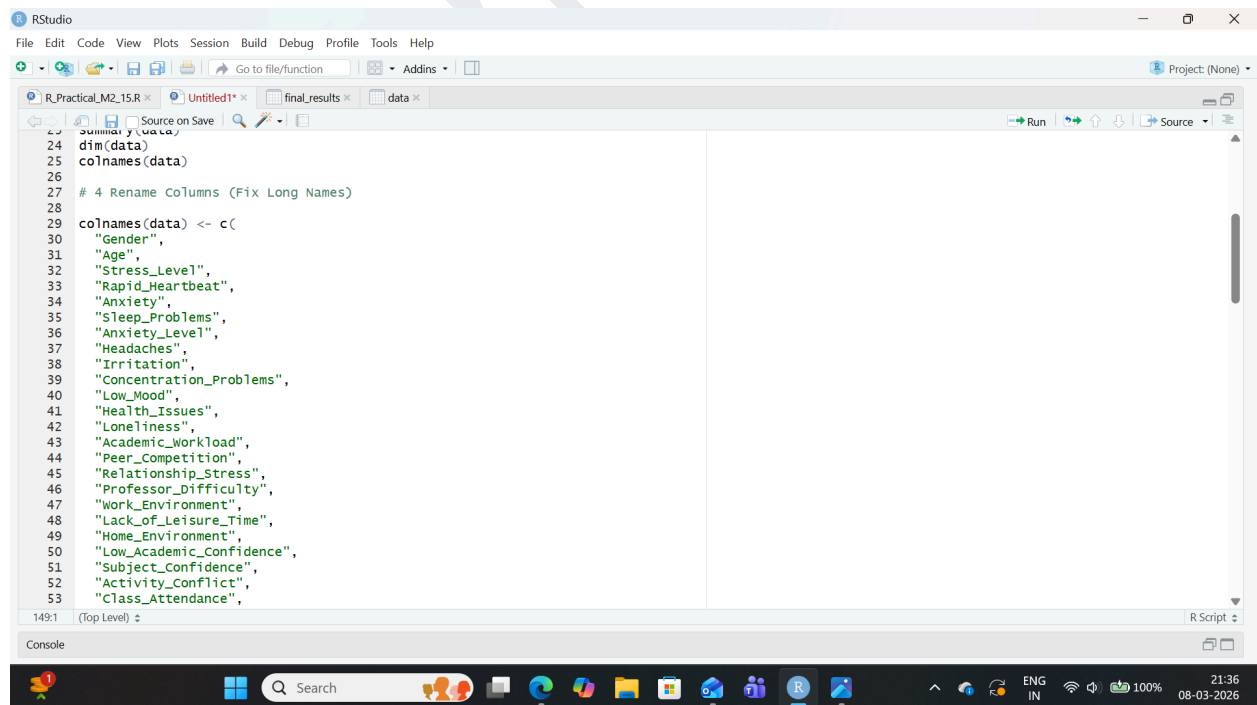


Implementation of Research using R

CODE :



```
1 # STUDENT STRESS DATASET ANALYSIS
2
3
4 # 1 Load Libraries
5
6 library(tidyverse)
7 library(ggplot2)
8 library(dplyr)
9 library(psych)
10 library(readr)
11
12 # 2 Import Dataset
13
14 data <- read.csv("student_stress.csv")
15
16 head(data)
17 tail(data)
18 View(data)
19
20 # 3 Dataset Information
21
22 str(data)
23 summary(data)
24 dim(data)
25 colnames(data)
26
27 # 4 Rename Columns (Fix Long Names)
28
29 colnames(data) <- c(
30   "Gender",
31   "Age",
32   "Stress_Level",
33   "Rapid_Heartbeat",
34   "Anxiety",
35   "Sleep_Problems",
36   "Anxiety_Level",
37   "Headaches",
38   "Irritation",
39   "Concentration_Problems",
40   "Low_Mood",
41   "Health_Issues",
42   "Loneliness",
43   "Academic_Workload",
44   "Peer_Competition",
45   "Relationship_Stress",
46   "Professor_Difficulty",
47   "Work_Environment",
48   "Lack_of_Leisure_Time",
49   "Home_Environment",
50   "Low_Academic_Confidence",
51   "Subject_Confidence",
52   "Activity_Conflict",
53   "Class_Attendance",
54 )
```

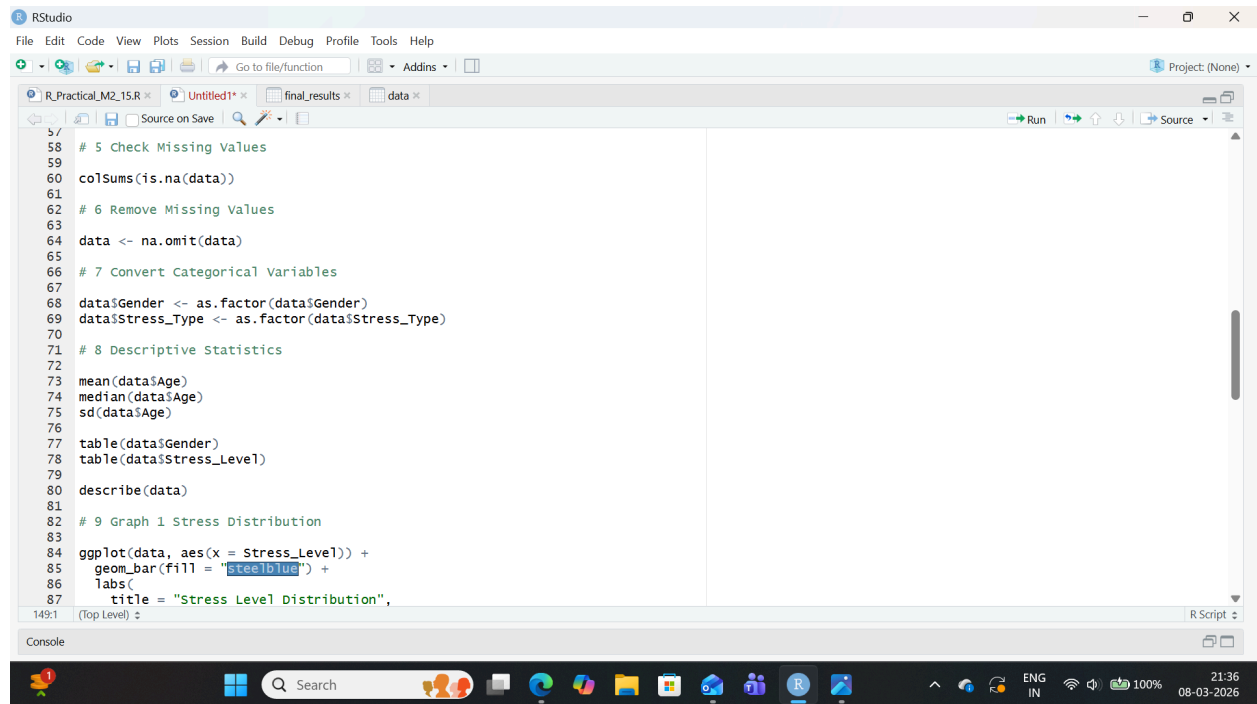


```
24 dim(data)
25 colnames(data)
26
27 # 4 Rename Columns (Fix Long Names)
28
29 colnames(data) <- c(
30   "Gender",
31   "Age",
32   "Stress_Level",
33   "Rapid_Heartbeat",
34   "Anxiety",
35   "Sleep_Problems",
36   "Anxiety_Level",
37   "Headaches",
38   "Irritation",
39   "Concentration_Problems",
40   "Low_Mood",
41   "Health_Issues",
42   "Loneliness",
43   "Academic_Workload",
44   "Peer_Competition",
45   "Relationship_Stress",
46   "Professor_Difficulty",
47   "Work_Environment",
48   "Lack_of_Leisure_Time",
49   "Home_Environment",
50   "Low_Academic_Confidence",
51   "Subject_Confidence",
52   "Activity_Conflict",
53   "Class_Attendance",
54 )
```

Name : Shubham Sanjay Karape

Role No : S085

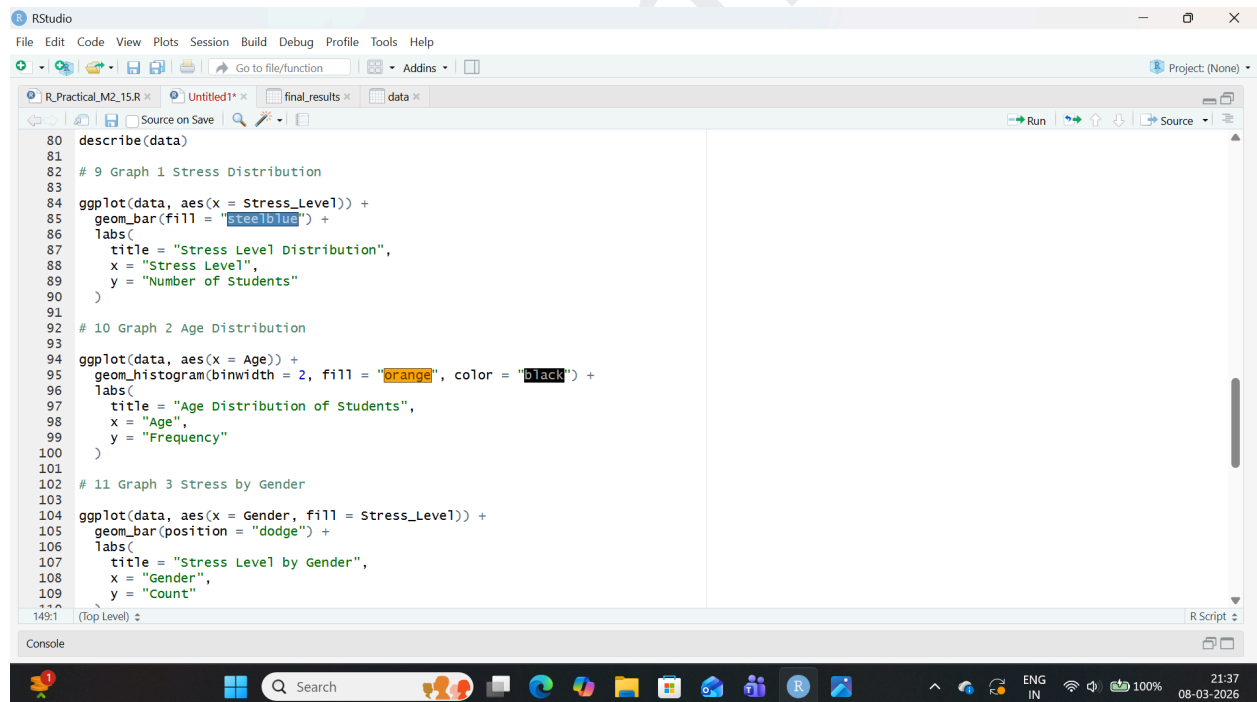
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The screenshot shows the RStudio interface with the following code in the script editor:

```
57 # 5 Check Missing Values
58
59 colSums(is.na(data))
60
61 # 6 Remove Missing Values
62
63 data <- na.omit(data)
64
65 # 7 Convert Categorical Variables
66
67 data$Gender <- as.factor(data$Gender)
68 data$Stress_Type <- as.factor(data$Stress_Type)
69
70 # 8 Descriptive Statistics
71
72 mean(data$Age)
73 median(data$Age)
74 sd(data$Age)
75
76 table(data$Gender)
77 table(data$Stress_Level)
78
79 describe(data)
80
81 # 9 Graph 1 Stress Distribution
82
83 ggplot(data, aes(x = Stress_Level)) +
84   geom_bar(fill = "steelblue") +
85   labs(
86     title = "Stress Level Distribution",
87     (Top Level)
88   )
```

The console is empty.



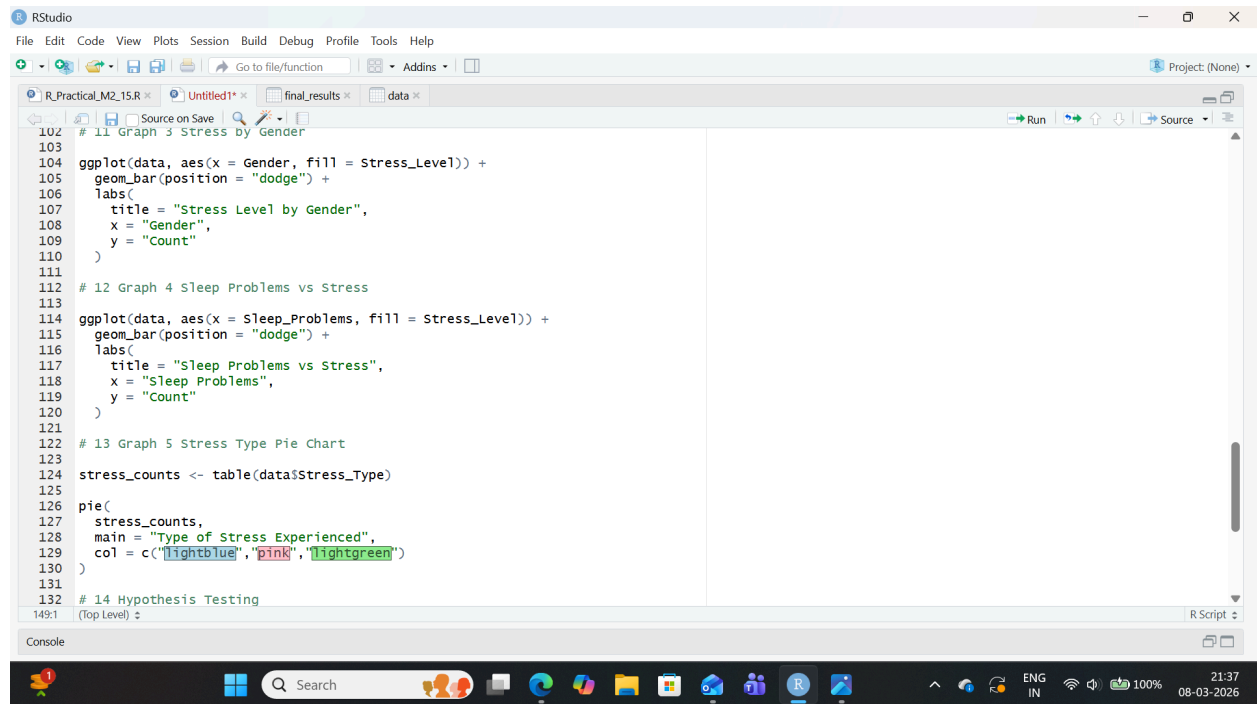
The screenshot shows the RStudio interface with the following code in the script editor:

```
80 describe(data)
81
82 # 9 Graph 1 Stress Distribution
83
84 ggplot(data, aes(x = Stress_Level)) +
85   geom_bar(fill = "steelblue") +
86   labs(
87     title = "Stress Level Distribution",
88     x = "Stress Level",
89     y = "Number of Students"
90   )
91
92 # 10 Graph 2 Age Distribution
93
94 ggplot(data, aes(x = Age)) +
95   geom_histogram(binwidth = 2, fill = "orange", color = "black") +
96   labs(
97     title = "Age Distribution of Students",
98     x = "Age",
99     y = "Frequency"
100  )
101
102 # 11 Graph 3 Stress by Gender
103
104 ggplot(data, aes(x = Gender, fill = Stress_Level)) +
105   geom_bar(position = "dodge") +
106   labs(
107     title = "Stress Level by Gender",
108     x = "Gender",
109     y = "Count"
110   )
111
112 (Top Level)
```

The console is empty.

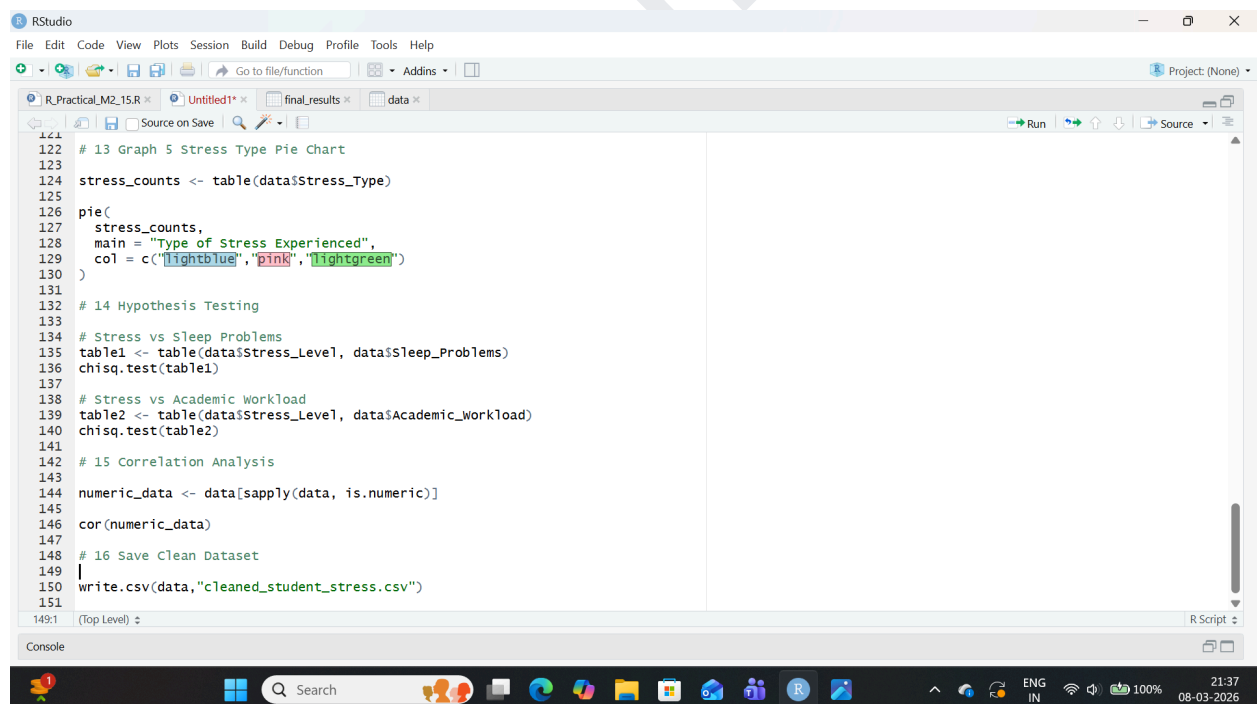
Name : Shubham Sanjay Karape
Role No : S085

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The screenshot shows the RStudio interface with a script editor containing R code. The code includes comments for three graphs: Graph 3 (Stress by Gender), Graph 4 (Sleep Problems vs Stress), and Graph 5 (Stress Type Pie Chart). The code uses ggplot2 for the bar charts and the pie() function for the pie chart. The console is empty.

```
102 # 11 Graph 3 Stress by Gender
103
104 ggplot(data, aes(x = Gender, fill = Stress_Level)) +
105   geom_bar(position = "dodge") +
106   labs(
107     title = "Stress Level by Gender",
108     x = "Gender",
109     y = "Count"
110   )
111
112 # 12 Graph 4 Sleep Problems vs Stress
113
114 ggplot(data, aes(x = Sleep_Problems, fill = Stress_Level)) +
115   geom_bar(position = "dodge") +
116   labs(
117     title = "Sleep Problems vs Stress",
118     x = "Sleep Problems",
119     y = "Count"
120   )
121
122 # 13 Graph 5 Stress Type Pie Chart
123
124 stress_counts <- table(data$Stress_Type)
125
126 pie(
127   stress_counts,
128   main = "Type of Stress Experienced",
129   col = c("lightblue", "pink", "lightgreen")
130 )
131
132 # 14 Hypothesis Testing
133 (Top Level)
```



The screenshot shows the RStudio interface with a script editor containing R code. The code includes comments for Graph 5 (Stress Type Pie Chart), Hypothesis Testing (Stress vs Sleep Problems and Stress vs Academic Workload), Correlation Analysis, and saving the dataset. The console is empty.

```
122 # 13 Graph 5 Stress Type Pie Chart
123
124 stress_counts <- table(data$Stress_Type)
125
126 pie(
127   stress_counts,
128   main = "Type of Stress Experienced",
129   col = c("lightblue", "pink", "lightgreen")
130 )
131
132 # 14 Hypothesis Testing
133
134 # Stress vs Sleep Problems
135 table1 <- table(data$Stress_Level, data$Sleep_Problems)
136 chisq.test(table1)
137
138 # Stress vs Academic Workload
139 table2 <- table(data$Stress_Level, data$Academic_Workload)
140 chisq.test(table2)
141
142 # 15 Correlation Analysis
143
144 numeric_data <- data[sapply(data, is.numeric)]
145
146 cor(numeric_data)
147
148 # 16 Save Clean Dataset
149
150 write.csv(data, "cleaned_student_stress.csv")
151
152 (Top Level)
```

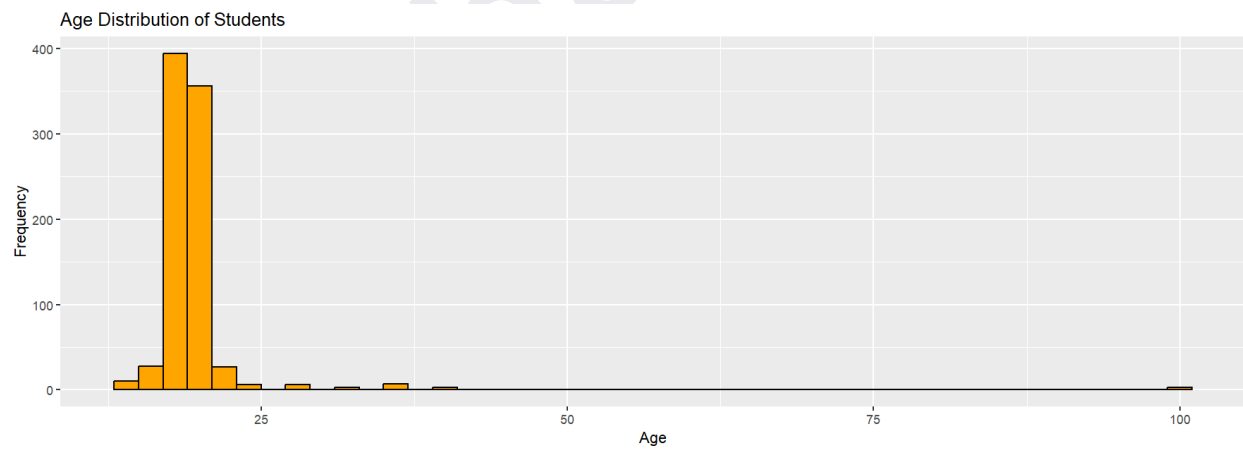
Name : Shubham Sanjay Karape
Role No : S085

GRRAPS :

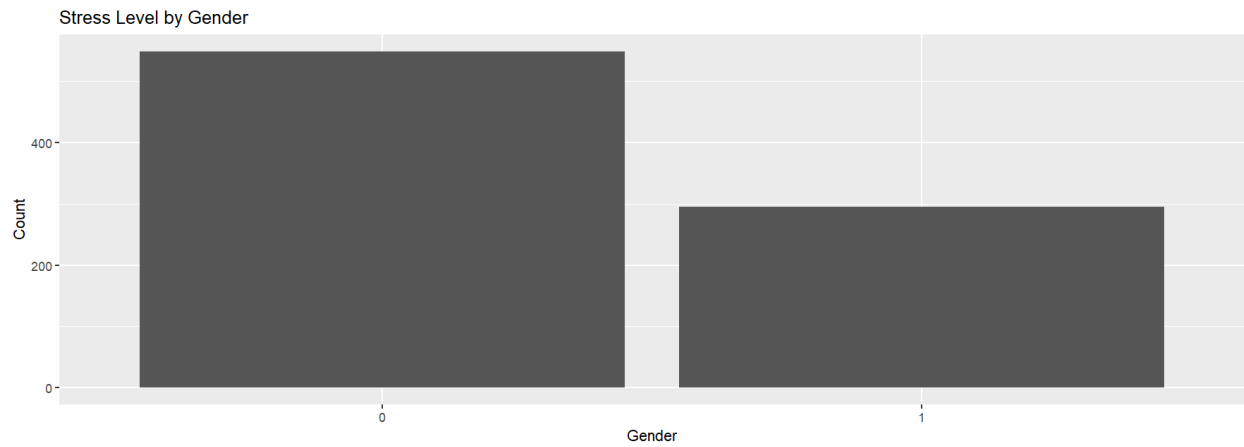
Stress Level Distribution



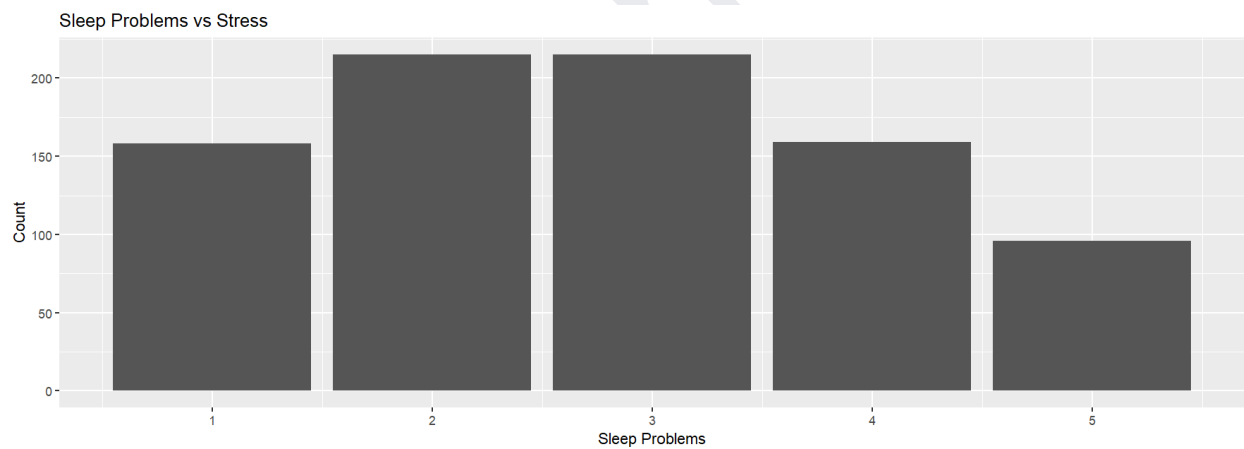
Age Distribution Of Students



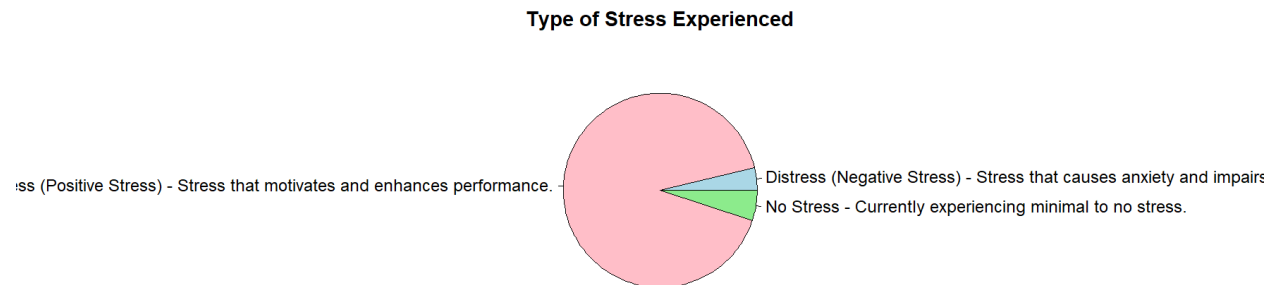
Stress Level By Gender



Sleep Problem Vs Stress



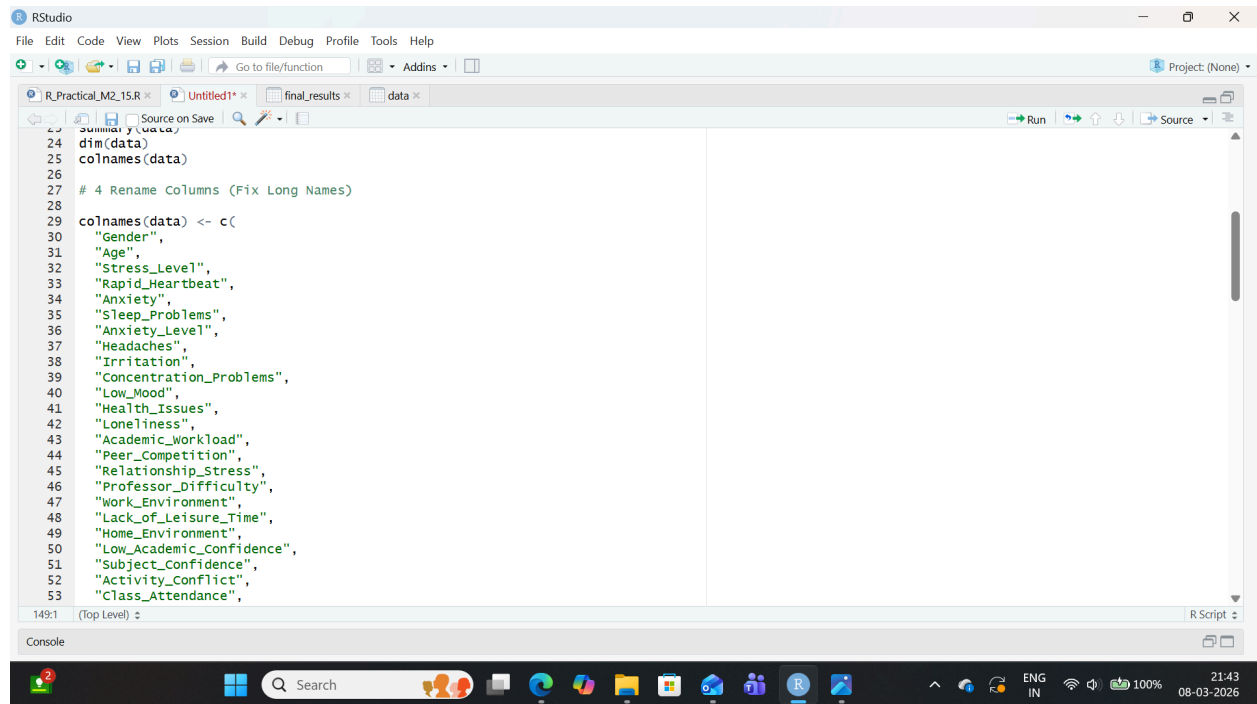
Types Of Stress Experienced



OUTPUT :

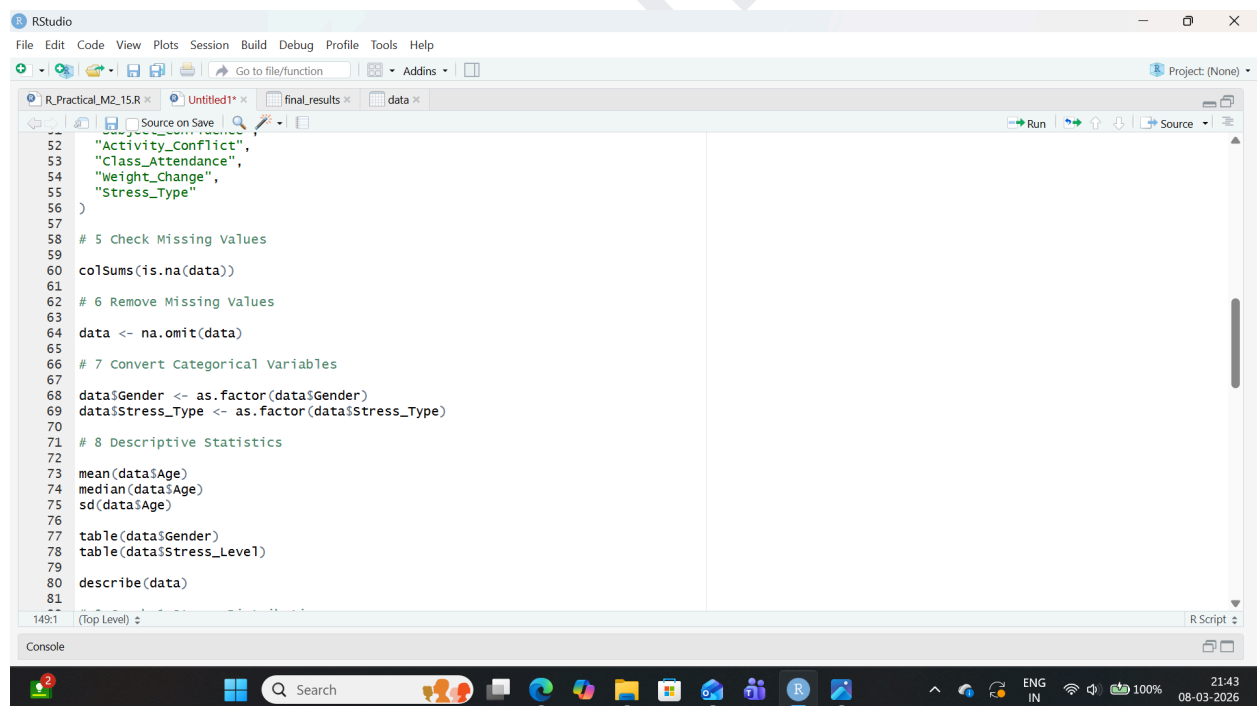
```
RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Addins Project: (None)
R_Practical_M2_15.R x Untitled1 x final_results x data x
Source on Save Run Source
1 # STUDENT STRESS DATASET ANALYSIS
2
3 # 1 Load Libraries
4 library(tidyverse)
5 library(ggplot2)
6 library(dplyr)
7 library(psych)
8 library(readr)
9
10 # 2 Import Dataset
11
12 data <- read.csv("student_stress.csv")
13
14 head(data)
15 tail(data)
16 View(data)
17
18 # 3 Dataset Information
19
20 str(data)
21 summary(data)
22 dim(data)
23 colnames(data)
24
25 # 4 Rename Columns (Fix Long Names)
26
27 colnames(data) <- c(
28   "Gender",
29   "Age"
30 )
31
149:1 (Top Level) R Script
Console
```

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The screenshot shows the RStudio interface with a script editor containing R code. The code defines a data frame, renames its columns, and lists the new column names. The console is empty.

```
24 dim(data)
25 colnames(data)
26
27 # 4 Rename columns (Fix Long Names)
28
29 colnames(data) <- c(
30   "Gender",
31   "Age",
32   "Stress_Level",
33   "Rapid_Heartbeat",
34   "Anxiety",
35   "Sleep_Problems",
36   "Anxiety_Level",
37   "Headaches",
38   "Irritation",
39   "Concentration_Problems",
40   "Low_Mood",
41   "Health_Issues",
42   "Loneliness",
43   "Academic_Workload",
44   "Peer_Competition",
45   "Relationship_Stress",
46   "Professor_Difficulty",
47   "Work_Environment",
48   "Lack_of_Leisure_Time",
49   "Home_Environment",
50   "Low_Academic_Confidence",
51   "Subject_Confidence",
52   "Activity_Conflict",
53   "Class_Attendance",
149:1 (Top Level)
```

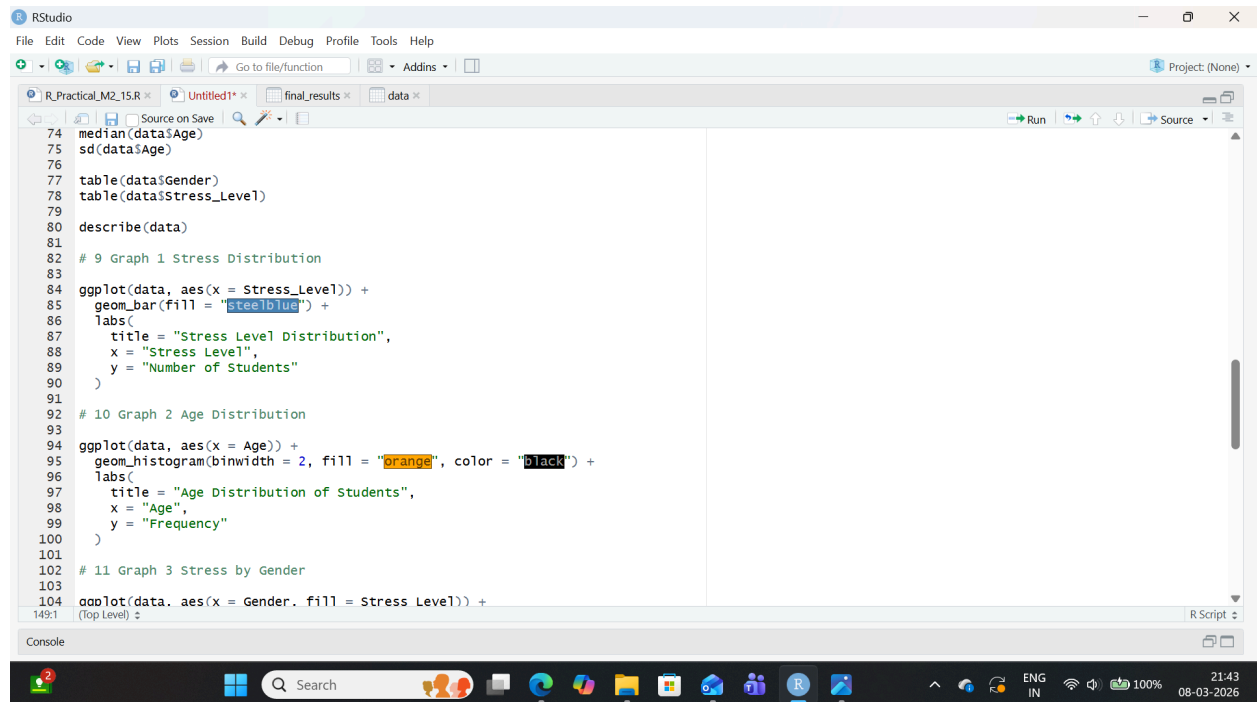


The screenshot shows the RStudio interface with a script editor containing R code for data cleaning and analysis. The code checks for missing values, removes them, converts categorical variables to factors, and calculates descriptive statistics. The console is empty.

```
52   "Activity_Conflict",
53   "Class_Attendance",
54   "Weight_Change",
55   "Stress_Type"
56 )
57
58 # 5 Check Missing Values
59
60 colSums(is.na(data))
61
62 # 6 Remove Missing Values
63
64 data <- na.omit(data)
65
66 # 7 Convert Categorical Variables
67
68 data$Gender <- as.factor(data$Gender)
69 data$Stress_Type <- as.factor(data$Stress_Type)
70
71 # 8 Descriptive Statistics
72
73 mean(data$Age)
74 median(data$Age)
75 sd(data$Age)
76
77 table(data$Gender)
78 table(data$Stress_Level)
79
80 describe(data)
81
149:1 (Top Level)
```

Name : Shubham Sanjay Karape
Role No : S085

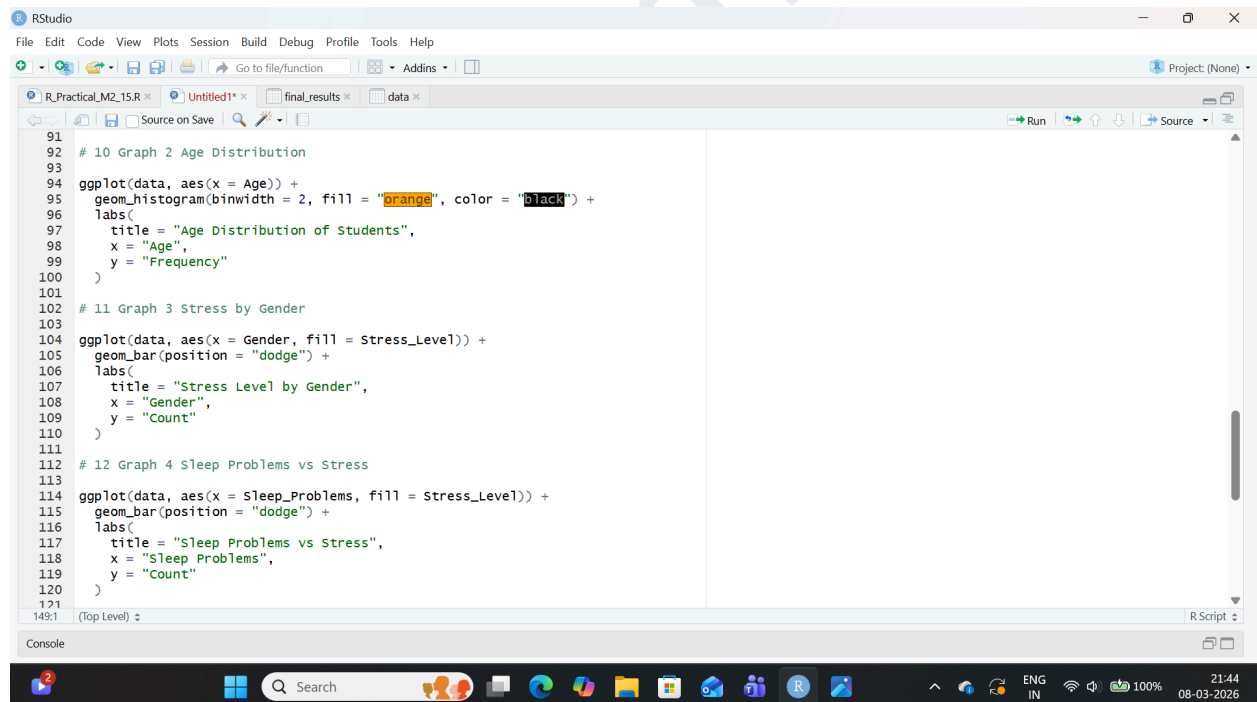
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The screenshot shows the RStudio interface with the following code in the script editor:

```
74 median(data$Age)
75 sd(data$Age)
76
77 table(data$Gender)
78 table(data$Stress_Level)
79
80 describe(data)
81
82 # 9 Graph 1 Stress Distribution
83
84 ggplot(data, aes(x = Stress_Level)) +
85   geom_bar(fill = "steelblue") +
86   labs(
87     title = "Stress Level Distribution",
88     x = "Stress Level",
89     y = "Number of Students"
90   )
91
92 # 10 Graph 2 Age Distribution
93
94 ggplot(data, aes(x = Age)) +
95   geom_histogram(binwidth = 2, fill = "orange", color = "black") +
96   labs(
97     title = "Age Distribution of Students",
98     x = "Age",
99     y = "Frequency"
100  )
101
102 # 11 Graph 3 Stress by Gender
103
104 ggplot(data, aes(x = Gender, fill = Stress_Level)) +
105   (Top Level)
```

The console is empty. The taskbar at the bottom shows the Windows Start button, a search bar, and various application icons. The system tray on the right indicates the language is set to English (IN), the battery is at 100%, and the date is 08-03-2026.



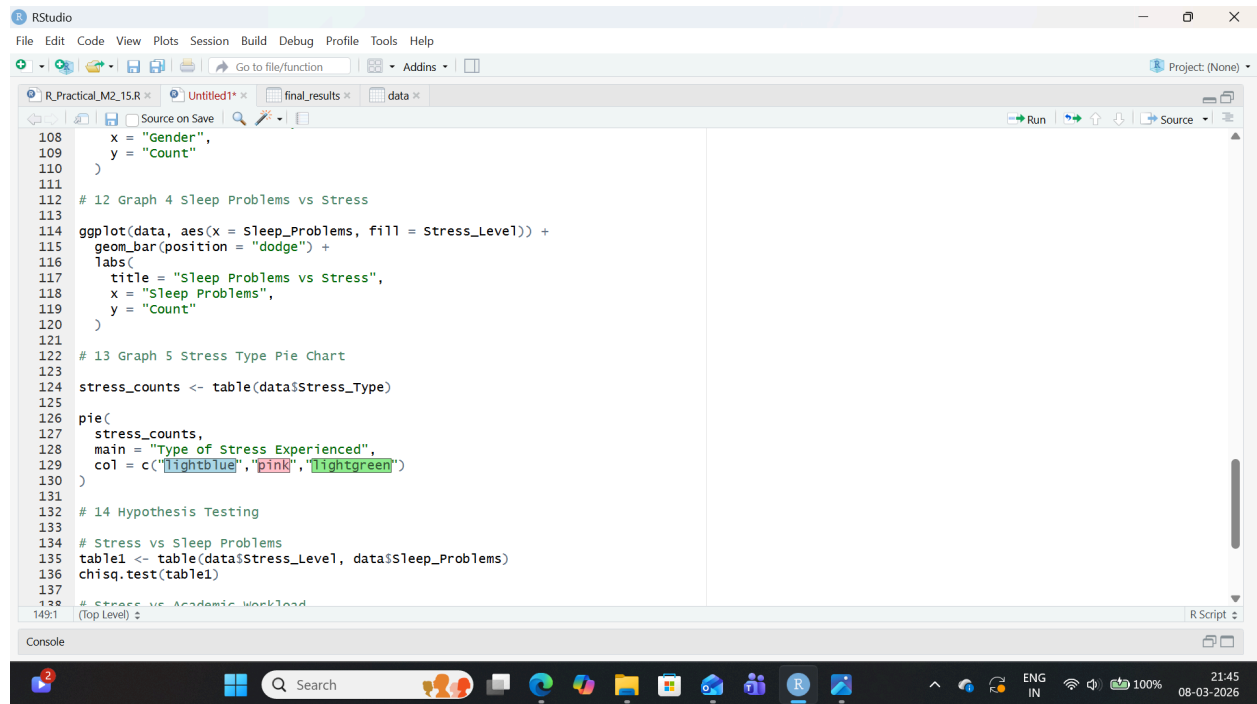
The screenshot shows the RStudio interface with the following code in the script editor:

```
91
92 # 10 Graph 2 Age Distribution
93
94 ggplot(data, aes(x = Age)) +
95   geom_histogram(binwidth = 2, fill = "orange", color = "black") +
96   labs(
97     title = "Age Distribution of Students",
98     x = "Age",
99     y = "Frequency"
100  )
101
102 # 11 Graph 3 Stress by Gender
103
104 ggplot(data, aes(x = Gender, fill = Stress_Level)) +
105   geom_bar(position = "dodge") +
106   labs(
107     title = "Stress Level by Gender",
108     x = "Gender",
109     y = "Count"
110   )
111
112 # 12 Graph 4 Sleep Problems vs Stress
113
114 ggplot(data, aes(x = Sleep_Problems, fill = Stress_Level)) +
115   geom_bar(position = "dodge") +
116   labs(
117     title = "Sleep Problems vs Stress",
118     x = "Sleep Problems",
119     y = "Count"
120   )
121
122 (Top Level)
```

The console is empty. The taskbar at the bottom shows the Windows Start button, a search bar, and various application icons. The system tray on the right indicates the language is set to English (IN), the battery is at 100%, and the date is 08-03-2026.

Name : Shubham Sanjay Karape
Role No : S085

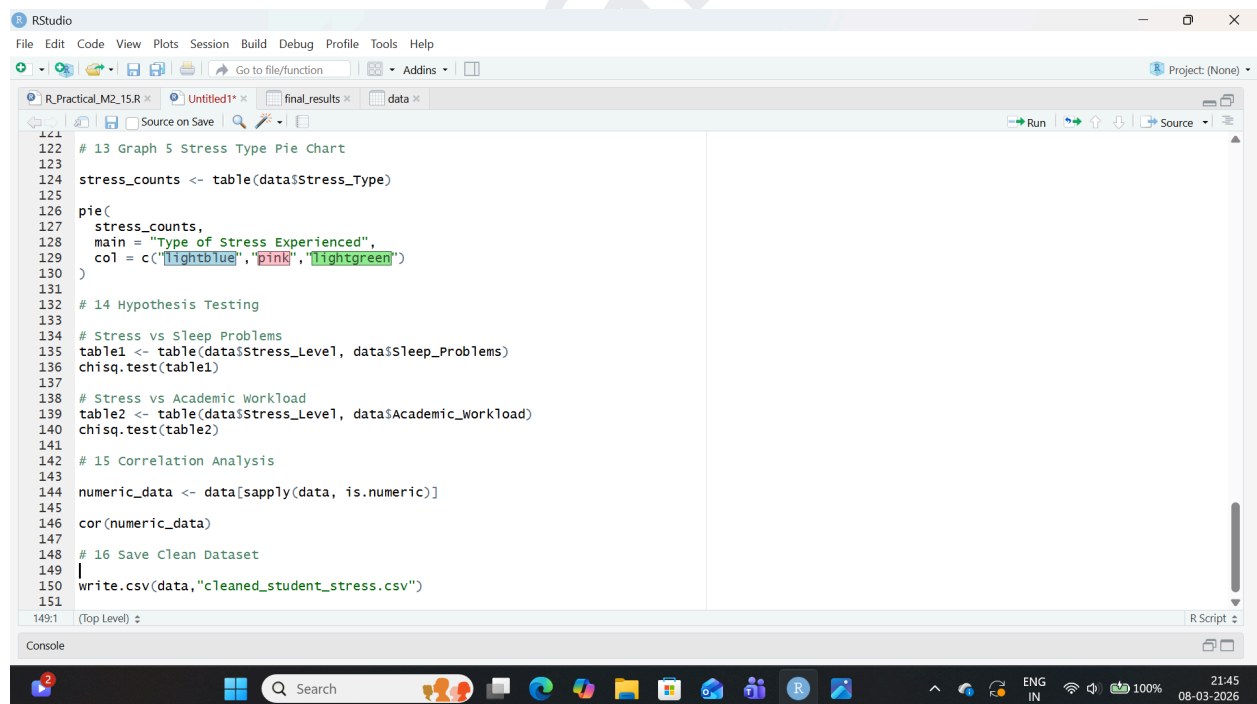
SHETH L.U.J. AND SIR M.V. COLLEGE OF ARTS, SCIENCE AND COMMERCE



The screenshot shows the RStudio interface with the following code in the script editor:

```
108   x = "Gender",
109   y = "Count"
110 )
111
112 # 12 Graph 4 Sleep Problems vs Stress
113
114 ggplot(data, aes(x = Sleep_Problems, fill = Stress_Level)) +
115   geom_bar(position = "dodge") +
116   labs(
117     title = "Sleep Problems vs Stress",
118     x = "Sleep Problems",
119     y = "Count"
120   )
121
122 # 13 Graph 5 Stress Type Pie Chart
123
124 stress_counts <- table(data$Stress_Type)
125
126 pie(
127   stress_counts,
128   main = "Type of Stress Experienced",
129   col = c("lightblue", "pink", "lightgreen")
130 )
131
132 # 14 Hypothesis Testing
133
134 # Stress vs Sleep Problems
135 table1 <- table(data$Stress_Level, data$Sleep_Problems)
136 chisq.test(table1)
137
138 # Stress vs Academic Workload
139 table2 <- table(data$Stress_Level, data$Academic_Workload)
140 chisq.test(table2)
```

The console is empty. The taskbar at the bottom shows the Windows Start button, a search bar, and various application icons. The system clock indicates 21:45 on 08-03-2026.



The screenshot shows the RStudio interface with the following code in the script editor:

```
122 # 13 Graph 5 Stress Type Pie Chart
123
124 stress_counts <- table(data$Stress_Type)
125
126 pie(
127   stress_counts,
128   main = "Type of Stress Experienced",
129   col = c("lightblue", "pink", "lightgreen")
130 )
131
132 # 14 Hypothesis Testing
133
134 # Stress vs Sleep Problems
135 table1 <- table(data$Stress_Level, data$Sleep_Problems)
136 chisq.test(table1)
137
138 # Stress vs Academic Workload
139 table2 <- table(data$Stress_Level, data$Academic_Workload)
140 chisq.test(table2)
141
142 # 15 Correlation Analysis
143
144 numeric_data <- data[sapply(data, is.numeric)]
145
146 cor(numeric_data)
147
148 # 16 Save Clean Dataset
149
150 write.csv(data, "cleaned_student_stress.csv")
151
```

The console is empty. The taskbar at the bottom shows the Windows Start button, a search bar, and various application icons. The system clock indicates 21:45 on 08-03-2026.

Name : Shubham Sanjay Karape

Role No : S085